

What Irregularity Costs: CUDA C++, Rust, and Triton on a Hash-Blocked GPU Workload

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Abstract

GPU language comparisons are almost always run on tiled dense linear algebra, where every toolchain is good and the differences are small. We implement the same hash-blocked TSDF fusion kernel five times, in CUDA C++, in Rust through NVIDIA’s `cuda-oxide`, and in Triton, and measure it on a workload with the opposite character: an open-addressed hash table with compare-exchange insertion, data-dependent per-lane probe depth, and contended scatter.

The result is a split. On the regular stage, which walks a truncation band and accumulates, all three languages land within a small factor of each other. On the irregular stage, which probes and inserts, Rust stays close to hand-written CUDA C++ while Triton is more than an order of magnitude slower. Language choice is nearly free on the work that is usually benchmarked and expensive on the work that is not.

We attribute both gaps to specific things the languages cannot express, rather than reporting ratios. Triton’s cost follows from a probe loop that must run to a compile-time bound and from `tl.atomic_cas` taking no mask, which forces a scratch structure with no counterpart in the CUDA implementation; sizing that structure badly also silently destroys the kernel’s ability to use additional streaming multiprocessors, a failure invisible on a single device. Rust’s cost was not visible in any instruction count: the Rust kernel issues fewer instructions, fewer compare-exchanges, and fewer registers at identical occupancy, and was still slower. Hardware counters located it in L1 residency. A GPU-scope atomic load must be coherent across SMs, no NVIDIA L1 is coherent across SMs, and so the type-correct way to read a shared location bypasses the cache on every access. Replacing two such reads with plain loads, correct because the algorithm rather than the memory model provides the guarantee, closes most of the gap.

We report the measurement discipline as a result in its own right. Roughly a dozen errors were caught during this work, three of which would have produced a reversed or null conclusion rather than a wrong magnitude, and two of which were caught only after the wrong number had been written down. We also report a defect found and fixed in `cuda-oxide` itself: its scoped atomic load and store could not be called at all in the build mode that produces real kernels.

1 Introduction

Pick up almost any comparison of GPU programming languages and the benchmark will be a matrix multiply, a convolution, a stencil, or a reduction. These are reasonable choices. They are also the workloads every toolchain has been tuned for, and they share a property that makes them easy: the work per thread is known before the kernel launches.

This paper measures the other case. A hash-blocked truncated signed distance function (TSDF) volume, the data structure underneath most real-time 3D reconstruction systems, is built by a kernel with none of that regularity. Each point walks a band of voxels, hashes each one to a

block coordinate, probes an open-addressed table for a variable number of steps, and either finds an existing block or claims one with a compare-exchange. Threads in the same warp do different amounts of work, contend for the same table slots, and must observe one another’s insertions.

We implement this kernel five times. Two of the implementations are CPU and third-party baselines used to establish correctness; three are the comparison proper: CUDA C++, Rust compiled through NVIDIA’s `cuda-oxide`, and Triton. All three operate on identical device memory with an identical hash layout, so what varies between them is the source language and nothing else.

Contributions.

1. A measured split between regular and irregular work (Section 5): the three languages are close on the stage that walks and accumulates, and separated by more than an order of magnitude on the stage that probes and inserts.
2. Attribution of both gaps to language-level causes (Section 6), including one that no instruction count reveals and that required hardware performance counters to find.
3. A catalogue of what each language cannot say (Section 7), catalogued from what the implementation work forced rather than from a wish list.
4. A measurement methodology and its failure modes (Section 4), reported because three of the errors it caught would have produced a confidently wrong conclusion rather than a noisy one.

What this paper is not. It is not a claim that one language is better. Rust lands within a few percent of hand-written CUDA C++ on real data once a single idiom is corrected, and Triton’s cost is concentrated in one stage that a practitioner could write in something else. It is a claim about where the costs are, and why.

2 The workload

A truncated signed distance function (TSDF) volume stores, at each voxel, the signed distance to the nearest observed surface, truncated to a band around it. Fusing a depth image means walking that band along each point’s view ray and accumulating a weighted average. The surface is later recovered as the zero crossing. It is the representation underneath KinectFusion and most real-time reconstruction systems since.

A dense volume is unaffordable at useful resolution, so implementations store only the blocks a surface actually touches, indexed by a spatial hash. That choice is what makes the kernel irregular, and it is why this workload is the vehicle for this paper rather than an application of it.

2.1 Structure

The volume is divided into blocks of $8^3 = 512$ voxels. A hash table maps block coordinates to indices in a flat pool of voxel data; five parallel float arrays hold the truncated distance, the accumulated weight, and three colour channels. The table is open-addressed with linear probing and sized to the next power of two above twice the pool capacity, so an occupied table is at most half full.

Each entry is 16 bytes: a 64-bit key and a 32-bit block index, with four bytes of padding. The key packs the three block coordinates into one word, each axis biased by 2^{20} into 21 bits:

$$\text{key}(x, y, z) = (x + 2^{20}) \ll 42 \mid (y + 2^{20}) \ll 21 \mid (z + 2^{20}).$$

Packing is not a space optimisation. It is what allows a single compare-exchange to publish an entire coordinate. An earlier layout stored the three axes separately and published them with a compare-exchange on the first followed by plain stores to the other two; a reader could then match the first word, mismatch on the rest, and probe onward, which at GPU thread counts degenerates into a full-table scan per lookup. The 64-bit key removes that failure by construction, and every arm inherits it.

2.2 Two passes, and why

Integration runs as two kernels over the same points.

Allocate walks the truncation band for each point, computes the block coordinate of each voxel it touches, and ensures a block exists for it. This is the irregular stage: threads probe for a data-dependent number of steps, race for the same slots, and must observe one another’s insertions.

Update walks the same band and accumulates into blocks that now exist. This is the regular stage: a lookup that almost always hits on the first probe, followed by five atomic additions per voxel.

Accumulation stores weighted *sums* rather than running means. A running mean requires reading the current value, combining, and writing back, which no single atomic can make safe across five arrays; sums commute, so plain atomic additions suffice and the arms need not agree on a locking scheme. The mean is recovered at extraction.

Both passes apply the same occlusion test. Applying it in only one leaves blocks that are allocated but never receive a contribution, which does not change the extracted surface but does change the block count, and the block count is one of the cross-arm correctness checks.

2.3 The insert protocol

Listing 1 is the CUDA C++ insert, and is the shape every arm implements.

Listing 1: The insert protocol. Both the probe loop and the publication window shape the comparison in Section 6.

```
for (probe = 0; probe < size; ++probe) {
    entry = table[(slot + probe) & mask];
    key = entry.key; // plain, cacheable load
    if (key == want) { // block already exists
        idx = entry.block_idx;
        while (idx < 0) // wait for the winner to publish
            idx = *(volatile int*)&entry.block_idx;
        return idx;
    }
    if (key == EMPTY) {
        prev = atomicCAS(&entry.key, EMPTY, want);
        if (prev == EMPTY) { // we won the slot
            idx = atomicAdd(&block_count, 1);
            block_coord[idx] = {bx, by, bz};
            __threadfence(); // coords visible before the index
            atomicExch(&entry.block_idx, idx);
            return idx;
        }
        // lost the race; fall through and re-examine this slot
    }
}
```

Two features of this protocol drive everything that follows.

The probe is unbounded. It runs until it finds the key or an empty slot, so its trip count is data-dependent and varies per lane. Section 7 shows this is not expressible in Triton at any price.

Publication is two steps. The key is published by a compare-exchange and the index by a later store, so a reader can observe a key whose index is still absent and must wait. The window exists because a block’s *identity* is known before insertion while its *location* is assigned by a counter and known only after winning. Publishing both atomically would need the location first; the location is only yours if the exchange succeeds. That circularity, and two attempts to remove it, are discussed in Section 6.

2.4 What is excluded, and why

Surface extraction is not part of the comparison. It exists once, in CUDA C++, and all three GPU arms call it, so measuring it three times compares nothing. Making it a real comparison would mean writing marching tetrahedra twice more, and that work would not extend the argument: it is a regular kernel, and the comparison already contains one in the update stage.

Investigating the decision also corrected a number. The extraction stage had been reported at 1.825 ms and as several times the cost of the whole integrate path. Splitting the call showed that 82% of that measurement on one GPU and 97% on the other was a device-to-host transfer over PCIe links of differing width, not compute. The kernel and its allocation are 0.319 ms, comparable to integrate rather than several times it, and the striking cross-device difference was the motherboard.

2.5 Why this workload

The properties that make TSDF fusion awkward are exactly the ones absent from the benchmarks these languages are usually compared on:

- work per thread is data-dependent and not known at launch;
- threads in a warp diverge on both probe depth and outcome;
- correctness depends on compare-exchange and on one thread observing another’s publication;
- the access pattern is a scatter through a hash, with no tile structure to exploit.

None of this is exotic. It is what any GPU hash table, sparse structure builder, free-list allocator, or dynamic work queue looks like. The results in Section 5 should be read as applying to that class, not to reconstruction specifically.

3 Implementations

Five implementations exist. Three are the comparison; two are there to make the three trustworthy.

A1, Open3D. A third-party voxel block grid, used as an external reference for the algorithm’s shape. Not timed against the others.

A2, CPU C++. The same algorithm on the host, in C++20. Establishes that the reference results are not a property of any GPU implementation.

A3, CUDA C++. The performance reference.

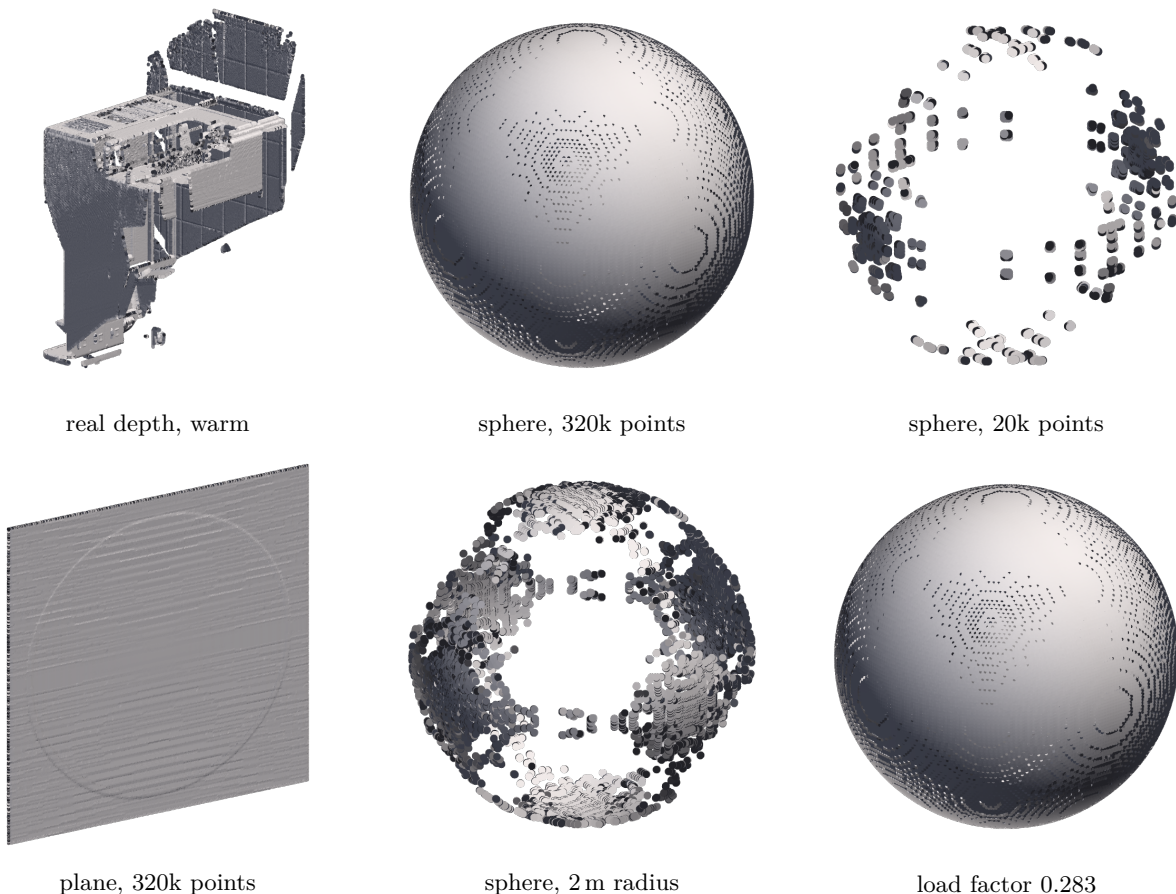


Figure 1: What the cells actually build, rendered from the fused volume: one lit point per occupied voxel, shaded by the surface normal the TSDF supplies. These are the reconstructions the timings come from, not illustrations of them. The 20k-point cell is the sweep’s outlier in every table, and the reason is visible here: at that density the volume barely reaches the weight threshold anywhere, so there is almost no work for the kernel to do and a fixed per-launch cost dominates whatever the language contributes.

A4, Rust. Compiled to PTX through `cuda-oxide`.

A5, Triton. Two variants: one where allocation runs on A3’s kernel and only the update stage is Triton, and one where both stages are Triton. The difference between them isolates the cost of the irregular stage without relying on cross-arm comparison.

3.1 What is held fixed

The three compared arms share the device memory layout exactly: the same 16-byte hash entry, the same packed key, the same multiply-xor hash, the same pool indexing, the same 256-thread launch geometry, and the same truncation band. Each arm is handed pointers into memory allocated once by the harness, through a C ABI, and none of them owns or reshapes it.

This is deliberate and it is a constraint on the result. Holding the layout fixed means the comparison measures how each language expresses one algorithm, not which algorithm each language makes convenient. A Triton implementation free to choose its own data structure would look different and probably better; that is a different paper, and a less controlled one.

The hash function in particular must be bit-identical across arms. A different mix changes probe sequences, which changes which blocks share cache lines, which would appear in the timings as a language difference.

3.2 What is not shared

No code is reused between arms. Each is written from the algorithm rather than ported from another arm’s source.

The original design had a vendored reference implementation that the other arms were ported from and then checked against. We removed it. Certifying an arm against the implementation it was derived from is weakly circular: a shared misunderstanding of the algorithm passes the check. The correctness design in Section 4 replaces it with closed-form geometry, which has no such failure mode.

3.3 Correctness

Every arm must reproduce a sphere of known radius to within a quarter of a voxel, agree with the reference on the number of allocated blocks exactly, and drop no points. In the final state the three GPU arms agree with each other to 0.000000000 m mean surface distance and 5.5×10^{-8} m Hausdorff distance, on 846,288 extracted vertices.

Timing is gated on those checks, and the sweep additionally discards any cell in which an arm drops a block or disagrees on block count. That gate is not ceremonial: it is what detected the silent data loss described in Section 7, in which one arm quietly built 13 fewer blocks than the others at a load factor no previous benchmark had reached.

3.4 Completeness

No arm is a sketch. Each implements the full algorithm: the truncation-band walk, the occlusion test, hash insertion with the publication protocol of Listing 1, pool-exhaustion reporting, and weighted accumulation across all five voxel arrays. Each passes the same correctness gate. The differences reported in Section 5 are therefore differences in how the same work is expressed, not in how much of it was done.

We deliberately do not compare line counts. The Rust source carries the measurement variants described in Section 6 alongside the production kernels, the Triton source carries two arm variants, and the CUDA source carries host-side management the others receive through the C ABI. Any line-count comparison across those would measure our file organisation rather than the languages.

4 Methodology

Most engineering benchmark papers are rejected for sloppy measurement rather than for uninteresting numbers. We describe the harness in detail, and then describe the errors it did *not* prevent, because the second list is more useful than the first.

4.1 Correctness gates timing

No arm is timed until it is known to compute the right answer, and the checks are ordered by how much they can catch.

Analytic first. Points are sampled on a sphere of known radius, so the extracted surface has a closed-form expected position. This is the only tier that can catch an error all arms share.

Third party second. Open3D’s voxel block grid provides an external reference for the algorithm’s shape.

Cross-arm last. The arms are compared to one another, at 0.000000000m mean surface distance in the final state.

The order is not decoration. A half-voxel bias, in which the signed distance was evaluated at the ray sample that selected a voxel rather than at the voxel centre, passed the plane test and every cross-arm comparison, because all arms had inherited it. It was caught only by the sphere, where it appeared as a uniform 0.005m radius error; correcting it improved agreement to 0.000015m, a factor of 340. Had the project been built on cross-arm agreement alone, every number in it would have been wrong in the same direction and the agreement would have looked like evidence.

4.2 What the harness defends against

Each of the following exists because its absence produced a wrong number at some point during the work.

Amortised launches. Each timed window contains several kernel launches rather than one, divided out afterwards. Timing a single launch measures the host’s wait for completion as much as the kernel: the driver spins briefly and then blocks, and the scheduler wakeup of roughly 130 μ s lands unevenly across arms depending on how much host work each does after enqueueing. This produced a spurious $4\times$ tail on one arm that a separate probe showed was entirely in the wait, with the kernel itself flat.

Allocation additionally needs distinct volumes per launch: after the first launch the table is populated, so re-launching into the same volume would measure a lookup-only fast path.

Interleaved and rotated arm order. Arms take turns within each repetition and the starting arm rotates. Running all of one arm and then all of another makes clock ramp indistinguishable from a language difference; a single session was observed going from 1575 MHz to 2842 MHz, which is larger than several of the effects being measured. A per-arm first-half against second-half comparison is printed as a check that the interleaving worked.

Exclusive device. The harness refuses to run when another process holds significant memory on the target GPU. An unrelated job sharing the device once made one arm measure 4.455 ms against its true 0.044 ms, a $100\times$ error that looked like a plausible result.

Distributions, not means. Median, 95th and 99th percentiles per stage, with the repeat count stated. Clocks and temperatures are recorded per cell rather than once per run, because a later cell running hotter than an earlier one is an artefact that is only visible if sampled alongside the timings.

4.3 The sweep

Every number comes from a matrix rather than a single scene, because a ratio measured once is a property of that scene’s contention pattern as much as of the language. Four axes, chosen so that each answers a stated question rather than filling a table: point count at fixed geometry, which raises contention while holding hash work constant; extent at fixed angular resolution, which is the opposite lever; hash load factor; and scene shape, including two cells of real depth data.

The matrix runs on two GPUs and three times over, and cells are discarded if any arm drops a block or disagrees with the reference on block count. Automatic reductions in batch size, when a cell’s volumes will not fit in memory, are reported rather than silently applied, because a cell measured at a different batch size is not comparable to one that was not.

4.4 What went wrong anyway

Roughly a dozen errors were caught during this work. Most produced a wrong magnitude. Three produced a reversed or null conclusion, and two of those survived long enough to be written up.

A workload asymmetry made the slower arm look faster. Colour accumulation was implemented in three arms and not the fourth, a 40% difference in atomic traffic. Rust appeared $2\times$ faster than CUDA C++.

A device axis was silently fake. One arm’s loader hardcodes device zero, and creating a driver-API context makes it current for the whole process, so an explicit device selection was ignored by every arm. The first two-device sweep reported a clean null result, identical performance across a $2.33\times$ difference in streaming multiprocessor count, while the second card sat idle at 8 W. That is a striking and publishable-looking finding with a ready explanation, and nothing in the timings looked anomalous. It was caught only because the null was surprising enough to prompt a check of the card’s power draw. The harness now selects the device by masking the process to one GPU and verifies the resulting context, because the runtime’s notion of the current device and the driver’s can differ.

A single pass was not reproducible. One cell reported 0.047 ms in one pass and 0.060 ms in each of three reruns, a 30% swing. No within-pass statistic separates that case: flagging on the ratio of the 95th percentile to the median fires on eight of ten cells and misses this one. Only repetition of the whole matrix detects it, which is why the sweep runs three passes and reports medians.

A fourth is worth listing because it was a stage-level rather than an arm-level error, and because it shows the same failure at a different granularity: the extraction stage’s reported time turned out to be mostly bus traffic rather than compute (Section 2).

Two more silent-measurement defects, found by widening the sweep. Adding real-data cells surfaced two failures of the same kind as the fake device axis, and we record them for the same reason. First, a cell that sizes its pool from the measured block count sized it from the frame alone, ignoring the warm pre-fill, so the pool was exhausted before the timed window opened and the cell measured pool exhaustion rather than table contention; because an unbounded probe against a full table scans every slot for every point, a cell that should run in two seconds ran for over half an hour, which is the only reason we noticed. Second, the Triton cubin path is resolved relative to the working directory, and the sweep driver runs from the build directory, so an override that is correct from the repository root silently drops the Triton arm: a complete sweep ran, reported “0 invalid”, and compared two arms where three were asked for. The harness now refuses to run with an arm missing unless told explicitly to allow it. A third instance, the same day, was a sweep run at the historical scratch-region size, which costs the Triton arm a factor of three to four; that value is no longer the default, because a default is what you measure by forgetting a flag.

4.5 What this suggests

The common thread is that all three reversing errors were failures of *comparability* rather than of precision. The two sides were not doing the same work, or were not running on the same machine, or were not being summarised by a statistic that survived repetition. None of them would have been caught by running more repetitions or reporting tighter error bars.

A language comparison’s credibility rests entirely on those three properties, and in this project each of them turned out to be something the harness had to verify rather than assume. We suggest that any such comparison state explicitly how it established them.

5 Results

Table 1 is the paper in one table.

Stage	Character	Rust / CUDA C++	Triton / CUDA C++
allocate	irregular: probe, CAS, publication	1.02–3.34	11.2–31.6
update	regular: walk and accumulate	0.96–1.19	1.1–2.6

Table 1: Ratio to hand-written CUDA C++ by stage, across every recorded cell of the sweep and both GPUs. The regular stage separates the languages by a small factor; the irregular stage separates them by more than an order of magnitude. The wide ends of both ranges are the smallest cell, where the kernel is short enough that launch effects dominate; see Section 5.

Workload	GPU	CUDA C++	Rust			Triton	
			ms	ms	×	ms	×
RetroOffice P000, warm	RTX 5070 Ti	0.388	0.396	1.02	1.132	2.92	
RetroOffice P000, warm	RTX 5060	0.763	0.768	1.01	2.229	2.92	
RetroOffice P000, cold	RTX 5070 Ti	0.472	0.486	1.03	1.215	2.57	
RetroOffice P000, cold	RTX 5060	0.905	0.915	1.01	2.335	2.58	
AmericanDiner P000, warm	RTX 5070 Ti	0.400	0.417	1.04	1.143	2.86	
AmericanDiner P000, warm	RTX 5060	0.780	0.796	1.02	2.105	2.70	
Plane, 320k points	RTX 5070 Ti	0.363	0.379	1.04	0.875	2.41	
Plane, 320k points	RTX 5060	0.673	0.686	1.02	1.665	2.47	
Sphere, 1.28M points	RTX 5070 Ti	0.975	1.023	1.05	2.982	3.06	
Sphere, 1.28M points	RTX 5060	1.709	1.737	1.02	5.707	3.34	
Sphere, 320k points	RTX 5070 Ti	0.278	0.308	1.11	0.828	2.98	
Sphere, 320k points	RTX 5060	0.511	0.538	1.05	1.526	2.99	

Table 2: Full integrate path, allocate plus update, median of three passes. Real depth data (TartanAir) and synthetic scenes, on two GPUs differing $2.33\times$ in streaming multiprocessor count.

5.1 The split

Figure 2 and Table 1 are the result. On the update stage, which walks a band and accumulates, the three languages differ by a small factor. On the allocate stage, which probes and inserts, Rust remains close to CUDA C++ while Triton is more than an order of magnitude behind.

The real-depth cells sit at the *worse* end of Triton’s range rather than the middle of it: across the nine valid real cells its allocate ratio spans $17.3\text{--}31.6\times$ with a median of $25.2\times$, against $11.2\text{--}31.6\times$ and a median of $21.5\times$ over the sweep as a whole. Rust does not show the same effect, spanning $1.12\text{--}1.41\times$ on real data against $1.02\text{--}3.34$ overall. Synthetic scenes therefore understate the cost of the construct Triton lacks, which matters because a synthetic scene is what a language comparison would ordinarily be run on.

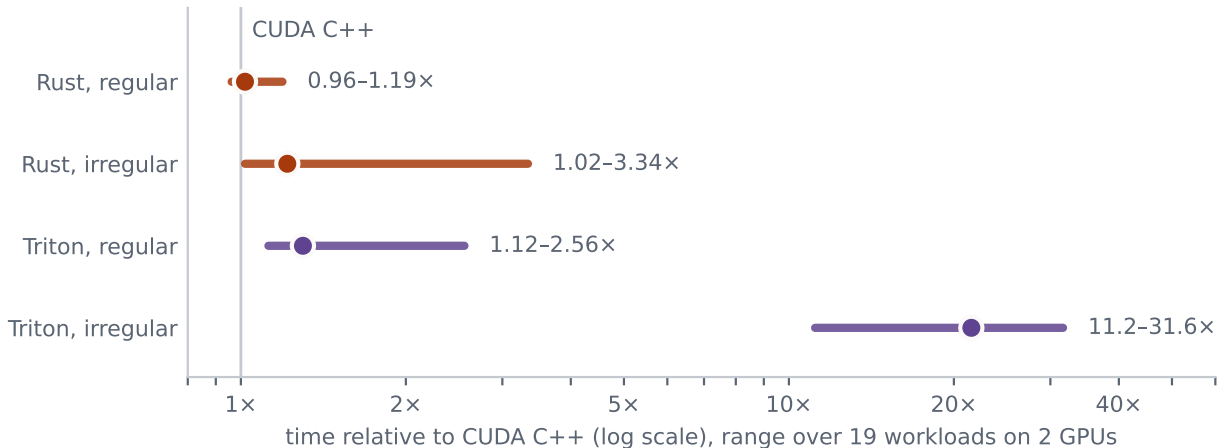


Figure 2: The result. The same two implementations, priced separately on the regular stage and the irregular one, as a range over all nineteen workload cells with the median marked. Both arms sit close to CUDA C++ on the stage that walks a band and accumulates; on the stage that probes and inserts, Triton is more than an order of magnitude behind and Rust’s cost becomes strongly workload-dependent. Log scale.

The wide ends of both ranges belong to the smallest cell, 20,000 points, where the kernel runs in a few microseconds and a fixed per-launch cost dominates anything the language does. We report the full range rather than excluding the cell, and Table 3 shows where it sits.

On the full integrate path (Table 2) the allocate stage is roughly a tenth of the work, so the totals are dominated by the regular stage: Rust lands within 1–3% of CUDA C++ on real depth data and Triton at 2.6–2.9 $\times$. A practitioner reading only the totals would conclude that Rust is free and Triton is expensive; reading the stages shows that both conclusions come from one kernel.

5.2 The sweep, and what each axis shows

Table 3 groups the cells by axis. Two of the axes were chosen as opposite levers, and they move the Triton ratio in opposite directions, which is the strongest evidence in the paper that the mechanism in Section 6 is the right one.

Point count, at fixed geometry. Raising the point count over the same surface raises contention while leaving the number of blocks unchanged. Triton’s ratio rises with it, from 14.2 $\times$ to 24.7 $\times$, because every additional lane that resolves early still issues a compare-exchange it cannot mask off. Rust’s ratio *falls*, from 3.34 $\times$ to 1.21 $\times$, which is the signature of a fixed cost being amortised rather than a multiplier: the absolute difference is roughly constant while the work grows 64 $\times$.

Extent, at fixed angular resolution. Spreading the same points over a larger sphere raises the number of blocks while lowering per-voxel contention. Triton’s ratio falls, from 21.3 $\times$ to 11.4 $\times$: with more blocks, fewer lanes resolve on the first probe, so the wasted work is a smaller share of the total. Two levers, opposite directions, one mechanism.

Load factor. The sharpest prediction of the Triton explanation is that its ratio should shrink as the table fills, because CUDA C++ pays actual probe depth while Triton pays a fixed bound. We

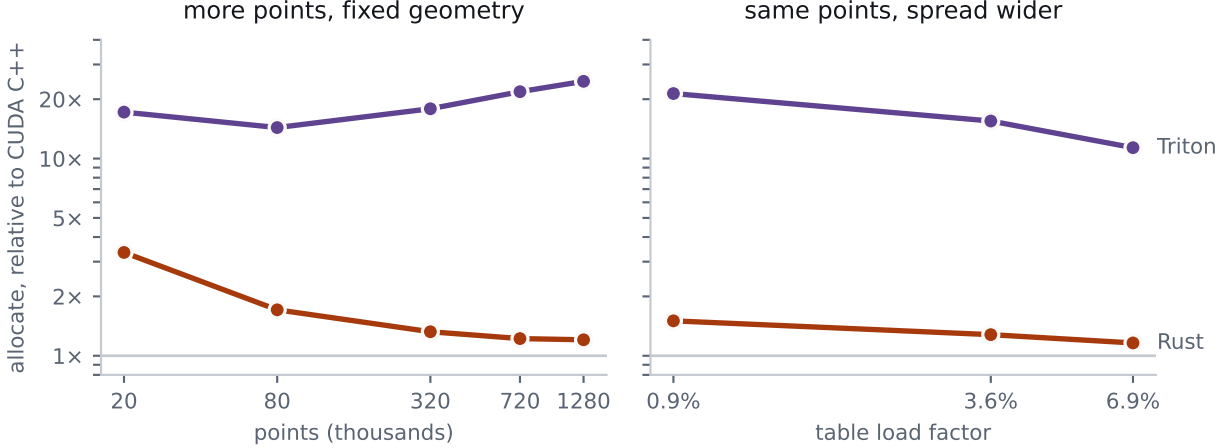


Figure 3: Two levers that move contention in opposite directions, on the RTX 5070 Ti. Adding points over fixed geometry raises the number of threads contending for each table slot, and Triton’s allocate cost rises with it; spreading the same points over more blocks lowers contention, and Triton’s cost falls. A cost driven by problem size would slope the same way in both panels. Rust falls in both, which is the shape of a fixed per-launch cost being amortised rather than a contention cost. Log-log.

could not test it. The table is sized at twice the pool and the pool must hold every block, so the reachable load factor caps at 0.283, over which range CUDA’s expected probe count only moves from 1.02 to 1.39. The lever is too weak, and this is a limitation of the data structure rather than of the sweep (Section 8).

Real data. Two cells unproject ground-truth depth from a TartanAir sequence. They validate the synthetic sweep rather than contradicting it: the ratios land inside the synthetic range and within a fraction of a point of the baseline cell. The warm cell, which integrates a frame into a volume already holding the preceding eight, is the regime a live pipeline is in for every frame after the first, and it is where Triton is worst in the entire sweep at $28.6\times$. In that regime most probes hit an existing key and exit immediately; CUDA C++ and Rust take that exit and Triton, which cannot, does not.

5.3 The device axis

The matrix runs on two GPUs differing $2.33\times$ in streaming multiprocessor count at identical code generation, since both are the same architecture. All three arms now scale with the machine, between $1.1\times$ and $2.2\times$ against an available $2.33\times$, with the low ends again the smallest cell.

This axis was not always uninteresting. Before a fix described in Section 6, Triton’s allocate did not scale at all, at $0.89\times$ to $1.07\times$: adding forty streaming multiprocessors bought it nothing, and on some cells the narrower card was faster. That failure is invisible on a single device, which reports a plausible time and no indication that the kernel has stopped responding to hardware.

The axis also serves as a check on the ratios themselves. Per cell, the two cards agree closely: Triton at $17.9\times$ on both at the baseline, Rust at 1.01 – $1.03\times$ on both with real data. A ratio that survives a change of machine balance is a better candidate for a language property than one that does not, and an earlier version of these numbers halved between the two cards, which was a

Cell	Axis	Points	Load	Rust	Triton
base	baseline	320,000	0.035	1.32	17.9
pts-20k	points	20,000	0.034	3.34	17.2
pts-80k	points	80,000	0.035	1.71	14.4
pts-720k	points	720,000	0.035	1.22	21.8
pts-1280k	points	1,280,000	0.035	1.21	24.6
r-0.25	extent	320,000	0.009	1.50	21.4
r-1.0	extent	320,000	0.036	1.28	15.5
r-2.0	extent	320,000	0.069	1.16	11.4
plane-320k	shape	320,356	0.032	1.21	17.1
lf-sparse	loadfactor	320,000	0.018	1.49	17.7
tartan-p001	real	374,692	0.043	1.33	31.6
diner-p000	real	393,624	0.094	1.33	26.4
tartan-p004	real	400,248	0.066	1.30	29.6
tartan-p005	real	401,678	0.108	1.30	24.8
tartan-p003	real	405,269	0.114	1.24	28.4
tartan-cold	real	407,444	0.078	1.23	17.8
tartan-warm	real	407,444	0.109	1.24	28.5
diner-p002	real	409,600	0.021	1.41	30.5
diner-p003	real	409,600	0.117	1.33	28.6

Table 3: Allocate ratio to CUDA C++ per cell, on the RTX 5070 Ti. Ratios, not times, because the axes change the absolute work by two orders of magnitude. The trends within each axis group are the evidence discussed below.

symptom rather than a finding.

6 Attribution

Reporting a ratio without a mechanism is what makes most language comparisons hard to cite. This section attributes both gaps, and reports the part of one that remains unattributed.

6.1 Triton: two costs that compound

The probe loop must run to a fixed bound. Triton has no per-lane early exit. A lane that resolves on the first probe cannot leave the loop; it can only be masked out of subsequent work while the loop continues. The bound is a `tl.static_range` trip count and therefore a compile-time constant, so every lane pays worst-case probe depth. Measured by varying the bound, the cost is linear in it: $8.38\times$ for an $8\times$ increase, confirmed again at a bound of 32.

`tl.atomic_cas` takes no mask. Unlike `tl.atomic_add`, the compare-exchange has no mask parameter, so lanes that have already resolved cannot be suppressed and still issue an exchange somewhere. They must be aimed at a scratch address holding a sentinel that can never satisfy the compare. The programmer thereby acquires a structure the CUDA version does not have, along with its sizing and its indexing.

The two compound: the bound sets how many iterations run, and the mask restriction sets what each wasted iteration costs. Replacing the exchange with a plain load, as a control, leaves

0.090 ms against 2.04 ms, a factor of 113, so essentially the entire gap is on this path rather than in code generation or memory layout.

6.2 Triton: a failure invisible on one device

The scratch region was originally indexed by lane, giving 256 addresses. Since lane indices repeat across the grid, *every* program used the same 256 addresses: the number of contenders grows with the grid while the address set does not.

The consequence was not primarily a slowdown. It was that the kernel stopped responding to hardware. Across a $2.33\times$ difference in streaming multiprocessor count, its allocate stage ran at $0.89\text{--}1.07\times$, and on some cells the *narrower* card was faster, which is the signature of a serialising address set: fewer contenders is marginally better. The other two arms tracked the machine at $1.8\text{--}2.1\times$ over the same cells.

Indexing the region per program instead restores scaling to $1.76\text{--}2.07\times$ and makes the kernel $3.5\times$ faster, from 1.812 ms to 0.524 ms. Saturation occurs before every lane has a private address, so contention is relieved statistically rather than eliminated.

This correction cost the headline. An earlier version of this work reported Triton’s allocate gap as $73\times$; sized correctly it is $17.9\times$. Most of the difference was a badly-chosen parameter of the workaround, not the language. We report the smaller number as the stronger claim, because it is device-independent where the larger one halved between cards.

The general lesson is the one a single-GPU benchmark cannot reach: the missing mask forces a structure whose bad settings are invisible in a throughput number. One GPU reports 1.8 ms and looks reasonable.

6.3 Rust: eleven hypotheses and a profiler

Counter	CUDA C++	Rust	Rust, fixed
static SASS instructions	520	528	528
branches (BRA/BSSY/BSYNC/BRX)	72	63	63
global memory ops (LDG/STG)	11	6	6
registers per thread	34	28	28
register spills	0	0	0
dynamic compare-exchange attempts	28,558	25,026	25,026
blocks built	1,160	1,160	1,160
achieved occupancy	84.30	84.27	84.27
active threads per warp	26.95	28.18	28.18
warp cycles per issued instruction	19.73	23.21	—
L1 sector hit rate	55.86	28.92	54.12
L2 sectors	568,655	964,189	552,775
long_scoreboard stall	12.02	15.63	11.06
duration	44.96	52.16	42.18

Table 4: Allocate kernel counters, 320k points, RTX 5070 Ti. Above the rule are counts of work: Rust does less on every one except static instruction count, where it is within 2%, and it was still slower. Below the rule are the counters that show why. “Rust, fixed” is after the change described in this section; the dash marks a figure not recorded for that build.

Rust’s allocate gap resisted attribution longer than any other result here, and the reason is visible in Table 4: it is not a *quantity-of-work* problem. The Rust kernel issues fewer branches,

fewer global memory operations, and fewer dynamic compare-exchanges, from a smaller register budget, at identical occupancy and with less warp divergence, producing an identical result. It does less work, more slowly.

Eleven hypotheses were tested and eliminated: register pressure, `clamp` semantics, volatile qualification, static instruction count, dynamic compare-exchange count, memory-ordering scope, the shared block counter, the publication fence, the publication store, the publication wait, and host-side launch overhead. Several were eliminated by experiments that required substantial work, including one that needed a patch to the Rust compiler backend before it could be run at all (Section 7).

Hardware counters located it in one run. Across every stall reason the profiler reports, `long_scoreboard`, the wait for a long-latency memory operation to return, was the *only* one that worsened, and it worsened by more than the whole gap; every other reason favoured Rust. The cause was L1 residency: identical requests over identical sectors, but $1.70\times$ the sectors pushed through to L2, and an L1 hit rate of 28.9% against 55.9%.

The mechanism. Two loads. The Rust implementation read the probe key and the published block index with a scoped atomic load; the CUDA implementation reads both plainly.

A note on sequence, because Section 7 reports that the scoped atomic load could not be called at all in the version of `cuda-oxide` we started from. It could not, and the Rust arm originally used a volatile read instead. That construct *also* bypasses L1, and more expensively, since Rust’s volatile semantics lower to a system-scope access. We patched the compiler in order to test a different hypothesis, which failed; the patch then made the scoped atomic load usable, and it is the version reported here. The relevant point survives the detour and is arguably strengthened by it: *both* constructs Rust offers for reading a location another thread writes bypass the cache, and the fix was to use neither. **A GPU-scope atomic load must be coherent across streaming multiprocessors, and no NVIDIA L1 cache is coherent across streaming multiprocessors, so such a load bypasses L1 by construction, on every call, however weak its ordering.** Relaxed removes ordering, not coherence.

The block index is almost always already published, so the Rust version paid an uncached access on the common path where the CUDA version pays a cached one and reserves the uncached access for when it is genuinely waiting. Making both first reads plain restores every counter to parity (Figure 4 and Table 4) and takes allocate from $1.63\times$ to $1.33\times$ at the baseline cell.

This is correct rather than a relaxation of correctness, and the distinction matters: the *algorithm* provides the guarantee that the memory model was being asked for. A stale key costs one extra probe; a stale index merely enters the wait loop, where an atomic load does observe the publication.

Why we think this generalises. The construct that is safe, idiomatic, and type-correct is the expensive one, and nothing in the source, the generated PTX, the machine code, the instruction counts, or the occupancy reveals it. On a GPU, “make this read safe against concurrent writers” and “make this read fast” are in tension in a way no type system expresses, because coherence and caching are the same mechanism. Any language that offers a scoped atomic load as the natural way to read shared state inherits this.

6.4 Two design experiments that failed

Both are reported because they bound the space of fixes.

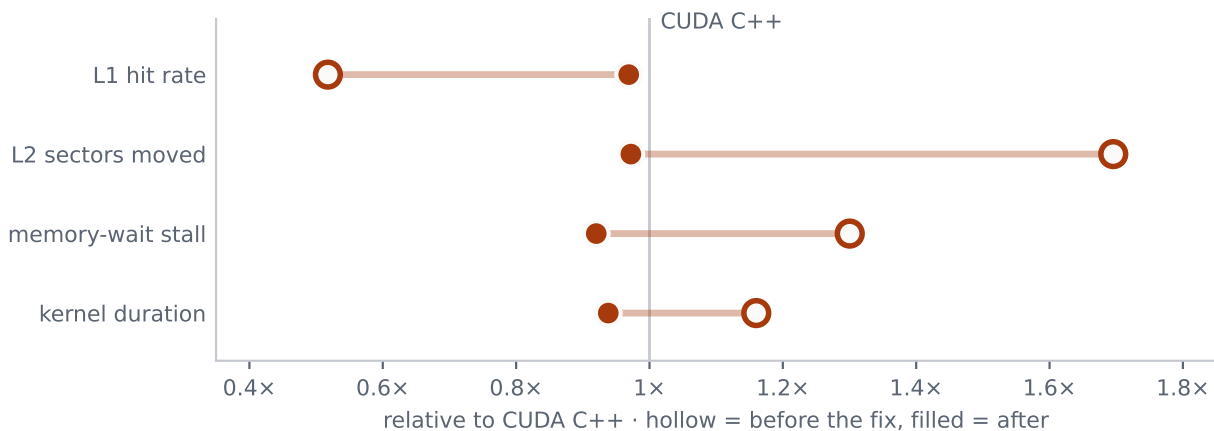


Figure 4: What the two-line change moved, each counter normalised to CUDA C++ so that quantities in different units share one axis. Hollow markers are the Rust kernel before the fix, filled markers after. The kernel was moving $1.70\times$ the L2 sectors from identical requests and retaining barely half as much in L1; afterwards every counter lands on the reference. Note that for L1 hit rate a value below 1 is the bad direction, since it is the only counter here where more is better.

Publishing key and index atomically. A 128-bit compare-exchange would publish both words at once and close the window described in Section 2, removing the wait. The instruction works in isolation; the algorithm does not. The location must be reserved before the exchange, so a thread that loses the slot holds one it cannot use, and at kernel start every thread targeting a block observes the same empty slot before any exchange lands: 65,536 reservations against 1,160 real blocks. This rules out the entire family of fixes that make the location part of the published value.

Deriving the location from the slot. Setting the block index equal to the hash slot removes the circularity properly, and is correct. It makes CUDA C++ 21–35% faster and Rust 29–32% faster, at the cost of sizing the voxel pool to the table rather than to the block count. It does *not* close the language gap: $1.64\times$ becomes $1.71\times$ at 320k points and $1.53\times$ becomes $1.39\times$ at 1.28M.

That last result corrected an earlier claim of ours. Removing the wait from the Rust arm alone had appeared to recover 75–83% of the gap, which we had reported as the wait being most of it. Removing it from both arms shows it is a large *shared* cost. The asymmetric measurement was the error, and it is the kind that only a symmetric re-run detects.

6.5 The residual

After the fix, Rust’s allocate stage spans $1.02\text{--}1.71\times$ across the sweep with a median of $1.21\times$, excluding the smallest cell, where the kernel is short enough that launch effects dominate and the ratio reaches $3.34\times$. That residual is not attributed. We report it as unattributed rather than assigning it to the nearest available cause, which over the course of this project would have been the wrong one eleven times running.

7 What each language cannot say

Section 6 priced what the languages cost. This section describes what they cannot say, which is what those prices are made of. Every item below was encountered while implementing the algorithm rather than constructed to make a point, and we separate genuine expressive limits from one defect that is merely a bug.

7.1 Triton: the probe loop

Listing 2 is the Triton probe, against the CUDA C++ of Listing 1.

Listing 2: The same probe in Triton. Both differences from Listing 1 are forced.

```
done = ~active
for p in tl.static_range(0, MAX_PROBE): # fixed trip count, no early exit
    slot = (start + p) & hash_mask
    eff = tl.where(done, my_scratch, slot) # resolved lanes aimed elsewhere
    key = tl.load(table_i64 + eff * 2)
    ...
    eff_cas = tl.where(empty, slot, my_scratch)
    old = tl.atomic_cas(table_i64 + eff_cas * 2, cmp, want) # no mask parameter
    done = done | won | hit
```

Two things are forced.

There is no per-lane early exit. Triton programs operate on blocks of lanes and the compiler owns the mapping from tensor elements to lanes. Allowing one element to leave a loop would tie control flow to that mapping and forfeit the compiler’s freedom to choose it, which is the freedom Triton’s performance on tiled workloads comes from. The restriction is principled. It is also why the loop above runs to a bound with a `done` mask.

The masks are being used outside their design. A `mask` on `tl.load` exists for ragged tile boundaries, where it suppresses a handful of lanes on one block out of many and its cost is negligible by construction. Carrying a `done` mask through a probe loop suppresses most lanes on most blocks for most iterations, and nothing in the cost model was built to make that cheap.

`tl.atomic_cas` takes no mask, and `tl.atomic_add` does. This asymmetry looks like drift rather than design: if masking an atomic were impossible, neither would have it. The consequence is that resolved lanes must be given somewhere harmless to write, so the programmer acquires a scratch region, its sizing, and its indexing, none of which appear in the CUDA version. Section 6 measured the cost of getting that sizing wrong at a factor of twenty, with the worst setting also destroying the kernel’s ability to use additional hardware.

7.2 Triton: an unbounded probe is inexpressible at any price

The bound is a `tl.static_range` trip count, so it must be a compile-time constant and the loop is unrolled. Raising it is not a workaround. Moving from 8 to 32 took roughly twenty minutes to compile against seconds, and produced a $3.7\times$ larger cubin.

The consequence is a correctness one. A bound small enough to compile is small enough to lose data, and the sweep now prices that across five load factors rather than the two we first reported:

Cell	Load factor	Blocks (CUDA C++)	Blocks (Triton)	Contributions dropped
lf-sparse	0.018	1,160	1,160	0
base	0.035	1,160	1,160	0
lf-lo	0.071	1,160	1,159	155
lf-mid	0.142	1,160	1,159	155
tartan-p002	0.192	25,230	25,230	22
lf-hi	0.283	1,160	1,147	27,943
tartan-dense	0.437	14,332	14,278	53,614

Every row reproduced identically in all six passes, so the loss is deterministic rather than a race. Two things in that table were not visible before. Loss begins at a load factor of 0.071, not at the 0.283 we previously reported as the first failing point; between 0.035 and 0.071 the arm goes from exact to losing a block. And `tartan-p002` is real depth data, at a load factor reached by an ordinary trajectory at the pool size every other real cell uses. We did not have to contrive that cell. It is one of six RetroOffice trajectories, and it is the one where the volume happens to be dense enough for a bounded probe to run out.

The failure is also not one thing. At 0.192 the block set is exact and 22 point contributions are discarded during the update stage, so the surface is complete and slightly wrong. From 0.283 upward whole blocks fail to be allocated and their surface is missing entirely. A consumer would notice the second and not the first.

It was also unreported. The implementation counted blocks lost to pool exhaustion, as the other arms do, but had no equivalent accounting for probe exhaustion, because the CUDA and Rust arms cannot exhaust a probe: they scan until they find a slot. We added the accounting, and note that its absence was the natural consequence of porting an algorithm whose failure mode does not exist in the original.

So the honest statement is stronger than a throughput ratio. *Triton cannot express an unbounded probe. The programmer must pick a bound. A bound small enough to compile is small enough to lose data at load factors CUDA C++ handles exactly, and the loss is silent, load-dependent, and absent from sparse benchmarks.*

7.3 Triton: diagnostics

Issuing an atomic on a scalar pointer rather than a tensor of pointers aborts the compiler with an MLIR assertion, **only integers and floats have a bitwidth**, and no source location. A compiler crash without a location is materially worse than a type error: it reports that something failed, not what the programmer did. We mention it because it is the kind of cost that never appears in a performance table and dominates the experience of writing the code.

7.4 Rust: the safe construct is the expensive one

Rust’s limit is not that something is inexpressible. It is that the natural, idiomatic, type-correct way to read a location another thread writes carries a guarantee stronger than the algorithm needs, and that guarantee has a fixed hardware cost.

A scoped atomic load is coherent at its scope. No NVIDIA L1 is coherent across streaming multiprocessors. Therefore the access cannot be served from L1, on every call, regardless of ordering: relaxed removes ordering, not coherence. Section 6 measured that at 27 points of L1 hit rate and $1.70\times$ the traffic to L2.

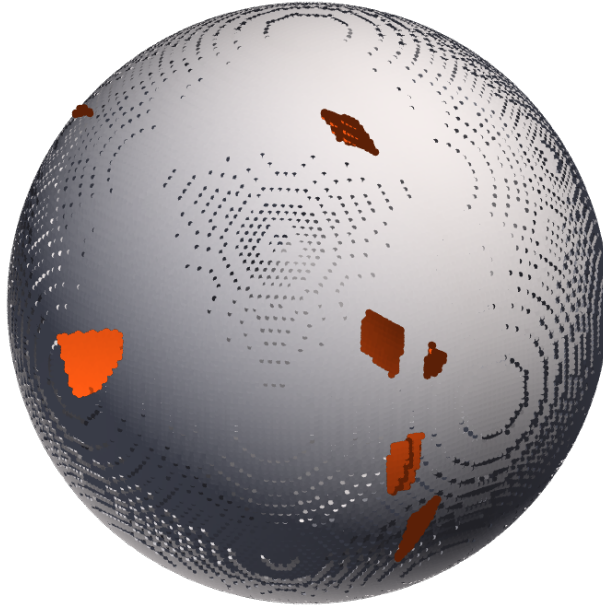


Figure 5: What that silence looks like. The surface is the volume CUDA C++ built at a load factor of 0.283; marked in colour are the points present there and absent from the volume Triton built from identical input. The loss is not scattered, which is the part a count does not convey: it arrives as whole contiguous patches, because the unit that fails to be allocated is a block of 8^3 voxels, and a block that loses its probe takes its entire surface with it. A reconstruction pipeline consuming this would see holes, not noise. On real depth the same failure has a different signature. At a load factor of 0.437 the `tartan-dense` cell loses 54 blocks scattered across the whole room rather than gathered into a few patches, because real occupancy is irregular where a sphere’s is not. That is the harder case to notice: pepper through a reconstruction reads as sensor noise, and a crater does not.

Nothing warns the programmer. The source looks correct, the generated PTX looks correct, the machine code looks correct, and the instruction counts favour the slower version. The cost is visible only in where the data is permitted to live.

The corresponding CUDA C++ is a plain load whose safety rests on an argument about the algorithm rather than on the type system: a stale key costs one extra probe, and a stale index enters a wait loop. That argument is not checkable by a compiler, which is exactly why the safer-looking construct is the one a conscientious programmer reaches for.

7.5 A defect, which is not the same thing

We separate the following from the above because it is a bug, and it has been fixed.

In the version of `cuda-oxide` used for this work, the scoped atomic load and store could not be called at all. Both lowered to LLVM IR forms that libNVVM rejects, so every call failed the backend verifier in the build mode that produces real kernels, with an error naming an LLVM basic block rather than the offending line of Rust. Any `AcqRel` or `SeqCst` ordering failed the same way through a rejected LLVM fence. The project’s own atomics example does not build in that mode.

The underlying PTX instructions have existed since `sm_70`, so this is a lowering gap rather than a hardware limit. We contributed a fix that lowers the three operations to those instructions, along with tests, a regression example, and the continuous-integration registration whose absence had allowed the defect to survive: the example that exercises these APIs was only ever run in the build mode where the problem does not appear.

This correction also revises a claim we made in an earlier draft. We had written that libNVVM’s restrictions force Rust back onto volatile reads and an explicit fence, and that this weakens the safety argument for Rust on GPUs. That is wrong. The scoped API exists precisely so callers need not do that; it was simply unusable. The distinction matters, because a design limitation and a bug support very different conclusions, and only one of them is an argument about the language.

8 Threats to validity

One architecture. Every measurement is on `sm_120`, across two GPUs differing $2.33\times$ in streaming multiprocessor count and roughly $2\times$ in memory bandwidth. A second architecture was considered and not run.

The magnitudes are therefore Blackwell figures and should be read as such. The mechanisms are not: Triton’s missing per-lane early exit and unmasked compare-exchange are properties of the language; a `tl.static_range` trip count being a compile-time constant is a property of the language; and a GPU-scope atomic load bypassing L1 follows from no NVIDIA L1 being coherent across streaming multiprocessors, which holds on every current architecture. We claim the ranking and its causes generalise, and that the numbers attached to them do not. Nothing in the argument rests on a magnitude.

The two-card axis is a partial mitigation rather than a substitute: it varies machine width at fixed code generation, which rules out the ratios being an artefact of one machine’s balance but says nothing about a different generation.

The sharpest prediction is untested. As Section 5 describes, the load factor reachable in this data structure caps at 0.283, which is too low to exercise the regime the Triton explanation makes its strongest prediction about. Testing it requires decoupling table size from pool capacity, which changes the shared structure and therefore every arm. The prediction is untested rather than unsupported.

An unattributed residual. After the fix in Section 6, Rust’s allocate stage remains $1.02\text{--}1.69\times$ CUDA C++ with a median of $1.21\times$, and we do not know why. Given that eleven prior hypotheses about this same gap were each plausible and each wrong, we prefer to leave it open than to assign it to whichever cause is nearest.

Two environments of real data. The real-data cells are nine valid cells from two TartanAir environments: six trajectories of RetroOffice and three of AmericanDiner. An earlier version of this work used one frame of one trajectory, and we flagged that as the weakest evidence in the paper. Widening it changed a conclusion rather than confirming one, which is the argument for having done it: real depth turns out to sit at the worse end of Triton’s range, not the middle, and one of the six RetroOffice trajectories reaches a load factor at which the Triton arm silently discards data. Neither was visible from a single scene. Two indoor environments from one simulator are still not a claim about real depth data in general, and in particular both are synthetic renderings

with exact ground-truth depth, so they have none of the noise, dropout or calibration error of a physical sensor.

We wrote all the implementations. Every arm was written by the same authors, who had substantially more experience in CUDA C++ than in Triton before this work. We mitigated this by holding the algorithm, memory layout, hash function and launch geometry identical across arms, by gating on correctness before timing, and by attributing every gap to a specific construct rather than resting on the totals. The Triton results in particular should be read as what a competent non-expert obtains from the documented API, and an expert may do better. We note that two of our own Triton and Rust implementations were materially improved during this work, in both cases by measurement rather than by insight, which suggests the remaining implementations may not be optimal either.

One workload family. All cells are TSDF fusion. The properties we claim to be measuring, data-dependent per-lane work, compare-exchange insertion and contended scatter, are shared with other GPU hash tables, sparse structure builders and dynamic work queues, but we have not measured those. The generalisation in Section 2 is an argument, not a result.

9 Related work

GPU programming model comparisons. Performance-portability studies compare CUDA, HIP, SYCL, Kokkos, RAJA and OpenMP across vendors, typically using proxy applications drawn from scientific computing [2]. Those proxies are dense and structured: stencils, particle updates, linear algebra. Our contribution is orthogonal rather than competing. We hold the architecture and the algorithm fixed and vary only the source language, and we choose a workload whose defining property is that the work per thread is not known at launch. To our knowledge no existing comparison targets that regime, which is also the regime in which we find the differences to be an order of magnitude rather than a few percent.

Triton. Triton [6] is built around tiles, and its central design decision is that the programmer describes what data a block needs while the compiler chooses how that data maps to lanes. Section 7 argues that the restrictions we encounter follow from that decision rather than from oversight, and that the workload we measure sits as far from its target as a GPU kernel can while remaining a GPU kernel. We intend the result as a boundary measurement for a tool that is very good inside its boundary, not as a criticism of the design.

Volumetric fusion. The truncated signed distance representation [1] and its real-time use [3] are long established, and the spatial hash that makes it affordable at scale is due to Nießner et al. [4]. We take the data structure from that line of work and contribute nothing to it; the reconstruction literature supplies our workload rather than our question. Open3D [8] provides the third-party implementation used as an external correctness reference, and the evaluation data are TartanAir V2 sequences, from the dataset introduced in [7].

Rust on GPUs. `cuda-oxide` [5] compiles Rust to PTX through libNVVM. Published comparisons against CUDA C++ are, as far as we can determine, absent: the project’s own comparison appendix is a placeholder. This paper is intended partly to fill that gap, and Section 7 reports a defect we found and fixed in the process.

Positioning. The closest work to ours in spirit is the performance-portability literature, which asks whether one source can run well everywhere. We ask a narrower question on harder ground: given one architecture and one algorithm, what does the choice of source language cost, and can that cost be attributed to something the language does or does not let you say.

10 Conclusion

We implemented one irregular GPU kernel in three languages and measured it on a matrix of workloads across two GPUs. The result is a split rather than a leaderboard. On the regular stage the languages differ by a small factor; on the irregular stage Rust stays close to hand-written CUDA C++ and Triton is more than an order of magnitude behind. Language choice is nearly free on the work that is usually benchmarked and expensive on the work that is not, which is an argument for benchmarking the other work.

Both gaps are attributable, and neither is attributable to code generation quality in the sense that phrase usually carries. Triton’s cost follows from a probe loop that must run to a compile-time bound and from a compare-exchange that cannot be masked; the second forces a data structure with no counterpart in the CUDA implementation, whose bad settings also stop the kernel responding to additional hardware while still reporting a plausible single-device time. Rust’s cost was invisible to every instruction count we could take, including counts that favoured Rust, and turned out to be that a GPU-scope atomic load cannot be served from L1 because coherence and caching are the same mechanism. The type-correct way to read shared state is the expensive one, and no artefact short of hardware counters shows it.

If there is a single transferable claim, it is that one. *On a GPU, making a read safe against concurrent writers and making it fast are in tension that no type system expresses.* Any language offering a scoped atomic load as the obvious way to read shared state inherits it, and the alternative is a plain load whose safety rests on an argument about the algorithm that a compiler cannot check. That is an uncomfortable place for a language whose appeal is that the compiler checks things.

We also report the measurement discipline, because three of the errors caught during this work would have produced a reversed or null conclusion rather than a noisy one, and two survived until after the wrong number had been written down. All three were failures of comparability rather than precision: the two sides were not doing the same work, or were not on the same machine, or were not summarised by a statistic that survived repetition. None would have been caught by more repetitions or tighter error bars. We suggest that comparisons of this kind state how they established those three properties, because in our case each was something the harness had to verify rather than assume.

Two results we would have preferred not to publish are included because omitting them would misrepresent how the work went. An earlier version of this paper reported Triton’s allocate gap as 73×; correctly configured it is 17.9×, and most of the difference was our own badly-sized workaround rather than the language. And we previously attributed most of Rust’s gap to the publication wait, on a measurement that removed that wait from one arm while leaving it in the other. The corrected figures are smaller and the corrected claims are narrower. They are also the ones that survived a symmetric test.

Future work. The load-factor regime where probe chains genuinely lengthen requires decoupling the table size from the pool capacity, and would test the sharpest prediction the Triton explanation makes. A second architecture would establish whether the magnitudes hold across a hardware generation; we expect the mechanisms to and the numbers not to. And the residual 1.02–1.69×

on Rust’s allocate stage remains open, which we note with some caution, having been confidently wrong about this particular gap eleven times already.

Artefacts. The harness, all five implementations, the raw measurement CSVs, and the scripts that generate every table in this paper are available. No number here was transcribed by hand.

Availability

The harness, all five implementations, the raw measurement CSVs from every sweep, and the scripts that generate each table in this paper are available with the source. Every table is produced from the CSVs by script; no measurement in this paper was transcribed by hand.

Disclosure of AI assistance

This work was carried out with substantial assistance from a large language model (Claude Opus 5, Anthropic). Specifically, the model wrote the five implementations, the measurement harness and the analysis scripts, executed the measurements, and produced the first draft of this manuscript, working interactively under the author’s direction across a single extended session.

The author defined the study, directed the investigation, and made the decisions that shaped it: which hypotheses to pursue, when to stop measuring and publish, what to exclude, and how to interpret the results. All measurements were run on the author’s hardware and the raw data are published so that any claim here can be checked independently of how it was produced. The author has reviewed the manuscript and is responsible for its content.

We note this workflow is visible in the paper’s record of corrections. Several figures reported in earlier drafts were later found to be wrong and are corrected in place rather than removed, including two in the conclusion. Those corrections came from measurement rather than from review, which is an argument for the methodology in Section 4 and not for the workflow.

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